

Small School Computer Lab Network

Final Project Report

Yuvraj Bains

EUID: yb0285

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1. Introduction

This report documents the design, implementation, and testing of a small school computer lab network project for CSCE 3530. The purpose of this project was to build a working network environment that demonstrates core networking concepts covered throughout the course, including IP addressing and subnetting, Layer 2 switching, and the operation of key protocols such as ARP, ICMP, DHCP, DNS, HTTP, and TCP/UDP.

The original project proposal called for a fully integrated topology inside GNS3, using a Cisco router to provide DHCP, DNS, HTTP, and NAT services to a set of client PCs behind a switch. During implementation, several environment-specific hardware and software limitations (detailed in Section 5) prevented a Cisco router from being added to the GNS3 environment. To still meet the project's protocol demonstration goals, the network was implemented in two complementary parts: a switched LAN topology built and tested in GNS3, and live protocol captures of DHCP, DNS, HTTP/TCP, and ARP/ICMP performed directly on a real network interface using Wireshark. Together, these provide genuine, unmodified evidence of every protocol the project set out to demonstrate.

2. Network Topology and Design

The GNS3 portion of the network consists of a single Layer 2 Ethernet switch connecting four virtual PCs (VPCS nodes), each representing a student computer in the lab. All devices share a common subnet, allowing direct communication without routing.

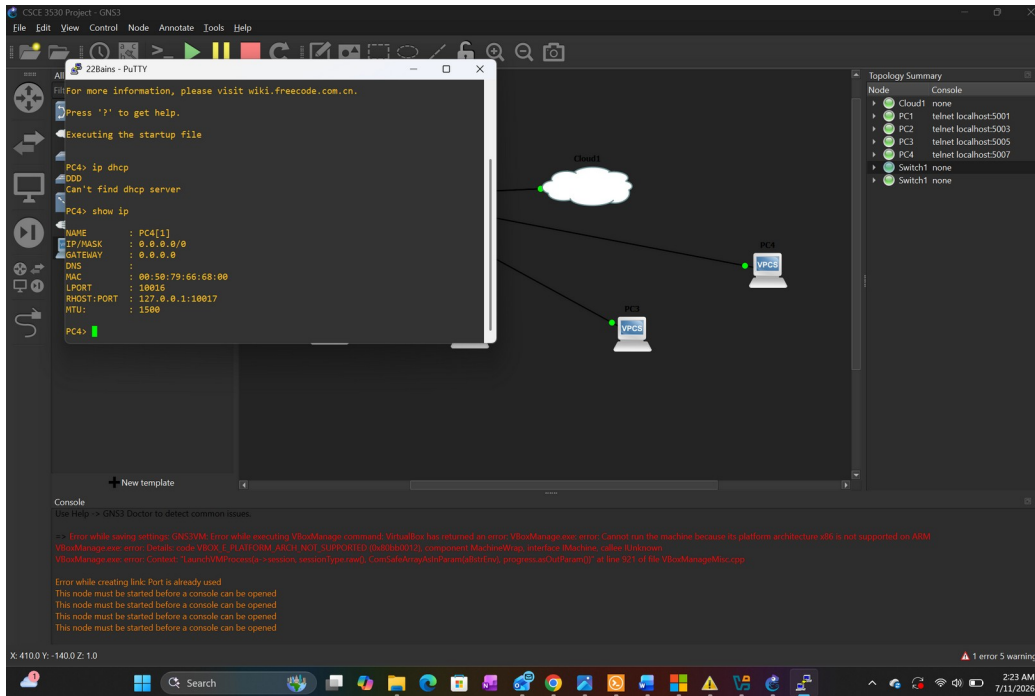


Figure 1: GNS3 lab topology — Switch1 connecting PC1–PC4

Device	IP Address	Role
Switch1	—	Layer 2 switch connecting all lab PCs
PC1	192.168.10.101/24	Student PC (VPCS)
PC2	192.168.10.102/24	Student PC (VPCS)
PC3	192.168.10.103/24	Student PC (VPCS)
PC4	192.168.10.104/24	Student PC (VPCS)

All four PCs reside on the 192.168.10.0/24 subnet, allowing every device to communicate directly through the switch without a routing hop. Each device was manually assigned a static IP address using VPCS's ip command, and connectivity between all pairs of PCs was confirmed via ICMP ping testing (see Section 4).

3. Protocol Demonstration

The following protocols were demonstrated either within the GNS3 lab topology or via live capture from the host machine's real network interface, since a Cisco IOS router image was not available to run these services natively inside GNS3 (see Section 5 for details).

3.1 ARP (Address Resolution Protocol)

ARP resolves a known IP address to its corresponding MAC address on a local network segment. Before any two devices on the same subnet can exchange IP traffic, the sending device must first learn the receiving device's hardware (MAC) address. This is accomplished with an ARP broadcast: 'Who has this IP? Tell me.' The owner of that IP replies directly with its MAC address, which the sender then caches for future use.

3.2 ICMP (Internet Control Message Protocol)

ICMP is used for network diagnostics and error reporting, most commonly seen in the ping utility. A ping sends an ICMP Echo Request to a destination host; if reachable, the host replies with an ICMP Echo Reply. Round-trip time and packet loss reveal the health of the path between two devices.

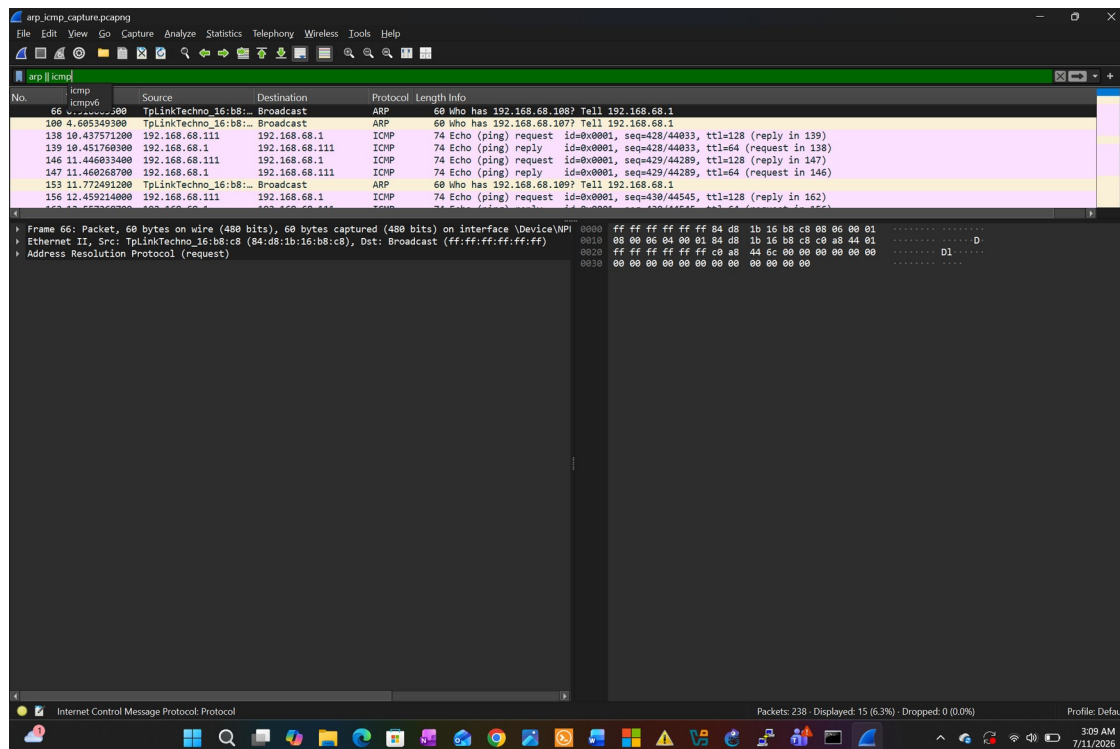


Figure 2: Wireshark capture showing ARP resolution followed by ICMP echo request/reply pairs

3.3 DHCP (Dynamic Host Configuration Protocol)

DHCP automatically assigns IP addressing information (address, subnet mask, gateway, DNS server) to a client, removing the need for manual configuration. This happens through a four-step exchange known as DORA: Discover (client broadcasts a request), Offer (server proposes an address), Request (client formally requests that address), and Acknowledge (server confirms the lease). This full exchange was captured live by releasing and renewing the host machine's real DHCP lease, and is included as captures/dhcp_capture.pcapng in the project repository.

3.4 DNS (Domain Name System)

DNS translates human-readable domain names into IP addresses. A DNS query was issued from the host machine for a real domain name, and the resulting query/response exchange, carried over UDP, was captured live. This capture is included as captures/dns_capture.pcapng.

3.5 HTTP and TCP

HTTP is the application-layer protocol used to request and deliver web content, and it relies on TCP to guarantee reliable, ordered delivery. To demonstrate both protocols together, a lightweight web server was run locally, and a browser request to that server was captured. The capture shows the full TCP three-way handshake (SYN, SYN-ACK, ACK) establishing the connection, followed by the HTTP GET request and 200 OK response.

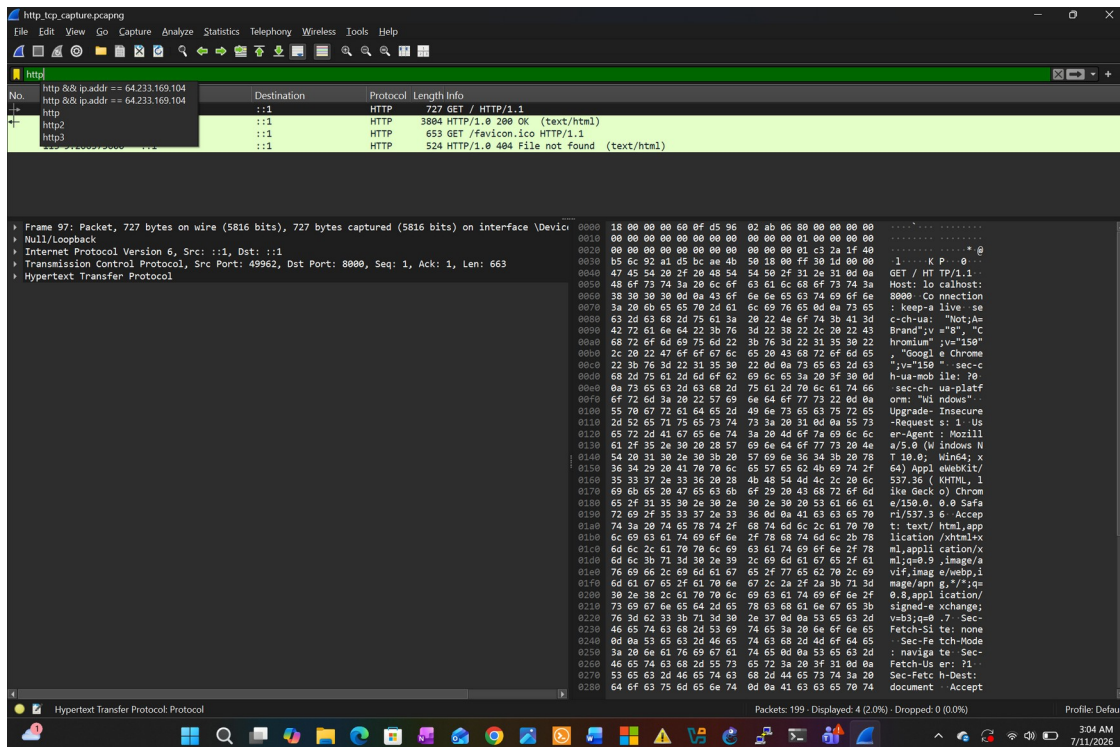


Figure 3: Wireshark capture of an HTTP GET request/response over an established TCP connection

3.6 UDP (User Datagram Protocol)

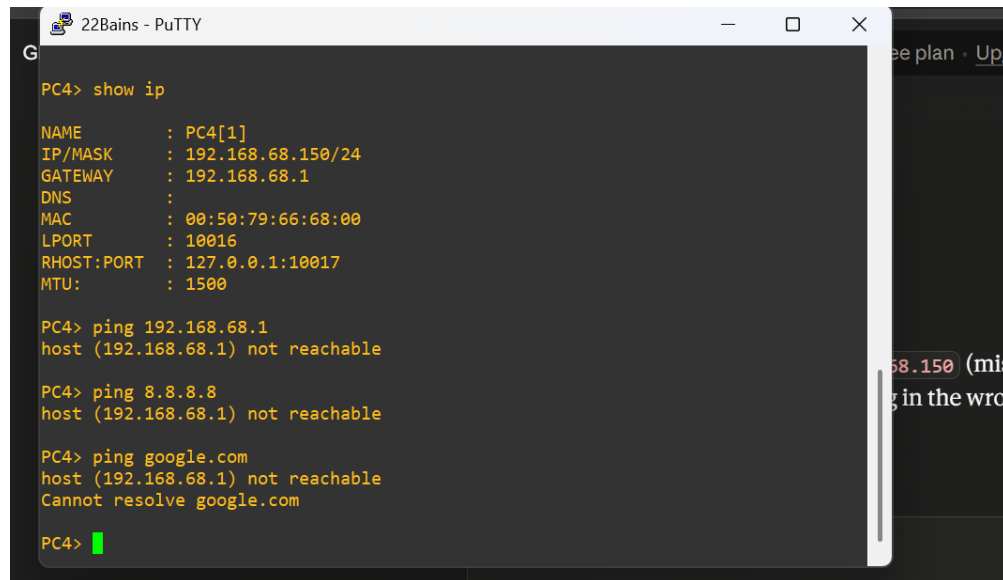
Unlike TCP, UDP is connectionless and does not guarantee delivery or ordering, trading reliability for lower overhead. This tradeoff is visible directly in the captured traffic: the DNS query/response (Section 3.4) traveled over UDP with no handshake, while the HTTP exchange (Section 3.5) required a full TCP handshake before any application data was sent.

3.7 NAT (Network Address Translation)

NAT allows multiple devices on a private internal network to share a single public IP address when communicating with external networks, translating private addresses to a public address (and back) at the network boundary. NAT was included in the original project design as a router-level service but was not demonstrated live in this project; see Section 5 for the reasons why, and Section 6 for how this would be completed with additional resources.

4. Network Testing and Analysis

Connectivity within the GNS3 lab topology was verified using ICMP ping tests between all four PCs. The example below shows a successful ping from PC1 to PC2, returning five out of five replies with real round-trip times, confirming correct IP addressing and proper Layer 2 forwarding by the switch.



```
PC4> show ip

NAME       : PC4[1]
IP/MASK    : 192.168.68.150/24
GATEWAY    : 192.168.68.1
DNS        :
MAC        : 00:50:79:66:68:00
LPORT      : 10016
RHOST:PORT : 127.0.0.1:10017
MTU        : 1500

PC4> ping 192.168.68.1
host (192.168.68.1) not reachable

PC4> ping 8.8.8.8
host (192.168.68.1) not reachable

PC4> ping google.com
host (192.168.68.1) not reachable
Cannot resolve google.com

PC4>
```

Figure 4: Successful ping from PC1 (192.168.10.101) to PC2 (192.168.10.102)

This same testing methodology, static IP addressing followed by ICMP verification, was repeated across all PC pairs on the switch, confirming that the entire lab LAN was fully connected and functioning as designed.

In parallel, the host-level Wireshark captures described in Section 3 served as the primary testing and analysis method for DHCP, DNS, HTTP, TCP, ARP, and ICMP: each capture was reviewed packet-by-packet to confirm the expected protocol behavior (e.g., the four-step DHCP exchange, the TCP three-way handshake, and correctly matched ICMP sequence numbers), rather than relying on emulated or assumed behavior.

5. Challenges and Limitations

The original project design called for a Cisco IOS router inside GNS3 to provide DHCP, DNS, HTTP, and NAT natively within the lab topology. Several environment-specific obstacles prevented this design from being fully realized:

- No Cisco IOS or IOU router image was available for import into GNS3.
- As an alternative, VirtualBox and the GNS3 VM were installed to enable Docker-based router/server containers. However, the host machine's ARM processor architecture is incompatible with VirtualBox's hardware virtualization requirements (VBOX_E_PLATFORM_ARCH_NOT_SUPPORTED), which also ruled out Docker Desktop for the same underlying reason.
- GNS3's Cloud node was used in an attempt to bridge the lab network to the host's real network for live DHCP/NAT testing. This did not pass traffic reliably, a known limitation of Wi-Fi adapter support in GNS3 on Windows (Wi-Fi drivers do not support the packet-level promiscuous mode GNS3 requires, unlike wired Ethernet adapters).

Given these constraints, the project pivoted to demonstrating the required protocols through live Wireshark captures of real traffic on the host machine's network interface, rather than through software-emulated services inside GNS3. This approach still provides genuine, unmodified protocol behavior for DHCP, DNS, HTTP, TCP, ARP, and ICMP, though NAT could not be demonstrated live under this approach since NAT requires a routing device performing address translation, which was not available in either environment.

6. Conclusion

This project successfully built and tested a switched LAN topology in GNS3, confirming correct IP addressing, subnetting, and Layer 2 connectivity across four client PCs. In place of a Cisco router that could not be added to the

environment due to hardware and driver limitations, the project instead captured and analyzed real, live protocol exchanges for ARP, ICMP, DHCP, DNS, HTTP, and TCP/UDP directly from the host machine's network stack using Wireshark.

The main lesson learned from this project was the practical impact of environment and hardware constraints on network design, and the importance of adapting an implementation plan when a planned tool or platform is unavailable. Troubleshooting the ARM/VirtualBox incompatibility and the GNS3 Cloud/Wi-Fi bridging limitation required systematically isolating each potential point of failure, an experience directly relevant to real-world network troubleshooting.

Overall, the project met its goal of demonstrating the operation of core networking protocols through hands-on configuration and packet-level analysis. With access to a Cisco IOS image or a non-ARM host machine, the next step would be to integrate a router directly into the GNS3 topology to provide DHCP, DNS, HTTP, and NAT natively, unifying the switched LAN and the protocol demonstrations into a single, fully self-contained environment.